

MAGNUM™ 3325 MT ABS Resin

Overview

MAGNUM™ 3325 MT is a resin with excellent processing characteristics and a matt surface finish. The mass (continuous process) ABS technology ensures an ABS resin that combines excellent processability with a stable light base colour that is ideal for self-colouring.

Applications

- Matt/unpainted interior automotive trim
- Door liners
- Dashboard components
- Pillar covers
- Consoles
- Glove boxes

Physical	Nominal Value (English)	Nominal Value (SI)	Test Method
Density			
--	1.04 g/cm ³	1.04 g/cm ³	ASTM D792
--	1.05 g/cm ³	1.05 g/cm ³	ISO 1183/B
Apparent Density	0.65 g/cm ³	0.65 g/cm ³	ISO 60
Melt Mass-Flow Rate (MFR)			
230°C/3.8 kg	2.5 g/10 min	2.5 g/10 min	ASTM D1238
220°C/10.0 kg	10 g/10 min	10 g/10 min	ISO 1133
Melt Volume-Flow Rate (MVR) (220°C/10.0 kg)	0.671 in ³ /10min	11.0 cm ³ /10min	ISO 1133
Molding Shrinkage			
Flow	4.0E-3 to 7.0E-3 in/in	0.40 to 0.70 %	ASTM D955
--	4.0E-3 to 7.0E-3 in/in	0.40 to 0.70 %	ISO 294-4
Mechanical	Nominal Value (English)	Nominal Value (SI)	Test Method
Tensile Modulus			
--	300000 psi	2070 MPa	ASTM D638 ¹
0.126 in (3.20 mm), Injection Molded	319000 psi	2200 MPa	ISO 527-2
Tensile Strength			
Yield	6130 psi	42.3 MPa	ASTM D638 ¹
Yield, 0.126 in (3.20 mm), Injection Molded	6240 psi	43.0 MPa	ISO 527-2/50
Tensile Strain			
Yield, 0.126 in (3.20 mm), Injection Molded	3.3 %	3.3 %	ISO 527-2/50
Break	25 %	25 %	ASTM D638 ¹
Flexural Modulus			
--	320000 psi	2210 MPa	ASTM D790 ²
0.126 in (3.20 mm), Injection Molded	305000 psi	2100 MPa	ISO 178 ^{3, 4}
Flexural Strength			
--	9350 psi	64.5 MPa	ASTM D790 ²
0.126 in (3.20 mm), Injection Molded	9430 psi	65.0 MPa	ISO 178 ^{3, 4}
Impact	Nominal Value (English)	Nominal Value (SI)	Test Method
Charpy Notched Impact Strength			
-22°F (-30°C), Compression Molded	3.8 ft-lb/in ²	8.0 kJ/m ²	ISO 179/2
-22°F (-30°C), Injection Molded	5.2 ft-lb/in ²	11 kJ/m ²	ISO 179/1eA
73°F (23°C), Injection Molded	8.1 ft-lb/in ²	17 kJ/m ²	ISO 179/1eA
73°F (23°C), Compression Molded	5.7 ft-lb/in ²	12 kJ/m ²	ISO 179/2

Impact	Nominal Value (English)	Nominal Value (SI)	Test Method
Notched Izod Impact			
73°F (23°C), 0.126 in (3.20 mm)	5.8 ft·lb/in	310 J/m	ASTM D256 ⁵
-22°F (-30°C)	4.8 ft·lb/in ²	10 kJ/m ²	ISO 180/1A
73°F (23°C)	8.1 ft·lb/in ²	17 kJ/m ²	ISO 180/1A
Instrumented Dart Impact			ASTM D3763 ⁶
-20°F (-29°C), 0.126 in (3.20 mm), Peak Energy	281 in·lb	31.7 J	
-20°F (-29°C), 0.126 in (3.20 mm), Total Energy	304 in·lb	34.3 J	
73°F (23°C), 0.126 in (3.20 mm), Peak Energy	275 in·lb	31.1 J	
73°F (23°C), 0.126 in (3.20 mm), Total Energy	402 in·lb	45.4 J	
Thermal	Nominal Value (English)	Nominal Value (SI)	Test Method
Deflection Temperature Under Load			
66 psi (0.45 MPa), Unannealed, 0.126 in (3.20 mm)	205 °F	96.1 °C	ASTM D648
264 psi (1.8 MPa), Unannealed, 0.126 in (3.20 mm)	181 °F	82.8 °C	ASTM D648
264 psi (1.8 MPa), Annealed	214 °F	101 °C	ISO 75-2/A
Vicat Softening Temperature			
--	227 °F	108 °C	ASTM D1525
--	216 °F	102 °C	ISO 306/B50
CLTE - Flow	5.2E-5 in/in/°F	9.4E-5 cm/cm/°C	ASTM D696
Flammability	Nominal Value (English)	Nominal Value (SI)	Test Method
Burning Rate (0.0787 in (2.00 mm))	2.4 in/min	60 mm/min	ISO 3795 ⁷
Flame Rating			UL 94 ⁷
0.0591 in (1.50 mm)	HB	HB	
0.118 in (3.00 mm)	HB	HB	
Carbon Emission	20.0 µg/g	20.0 µg/g	VDA 277 ⁷
Fogging (212°F (100°C))	98 %	98 %	ISO 6452 ⁷

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

¹ Type I, 2.0 in/min (51 mm/min)

² Type I, 0.20 in/min (5.1 mm/min)

³ 0.079 in/min (2.0 mm/min)

⁴ 3-points

⁵ 10 mil (0.25 mm) Notch Depth

⁶ 11.1 ft/sec (3.39 m/sec)

⁷ This rating not intended to reflect hazards presented by this or any other material under actual fire conditions.



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North America	
U.S.	+1-855-TRINSEO (+1-855-874-6736)
U.S. - Canada	+1-989-633-1718
Latin America	
Brazil	+55-11-5184-8722
Argentina, Chile, South Region of LAA	+54-11-4319-0100
Mexico, Colombia, North Region of LAA	+52-55-5201-4700
Europe/Middle East/Africa	
	+800-444-11-444
	+31-11567-2601
Asia Pacific	
China	+603-7965-53-19
	+86-21-3851-1017
Email	CIG@trinseo.com

www.trinseo.com

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